discrete r.v.

e.g., Bernoulli

Binomial

Poisson

Geometric

etc.

Continuous r.v.

e.g., Uniform

normal

chi-square

t-distribution

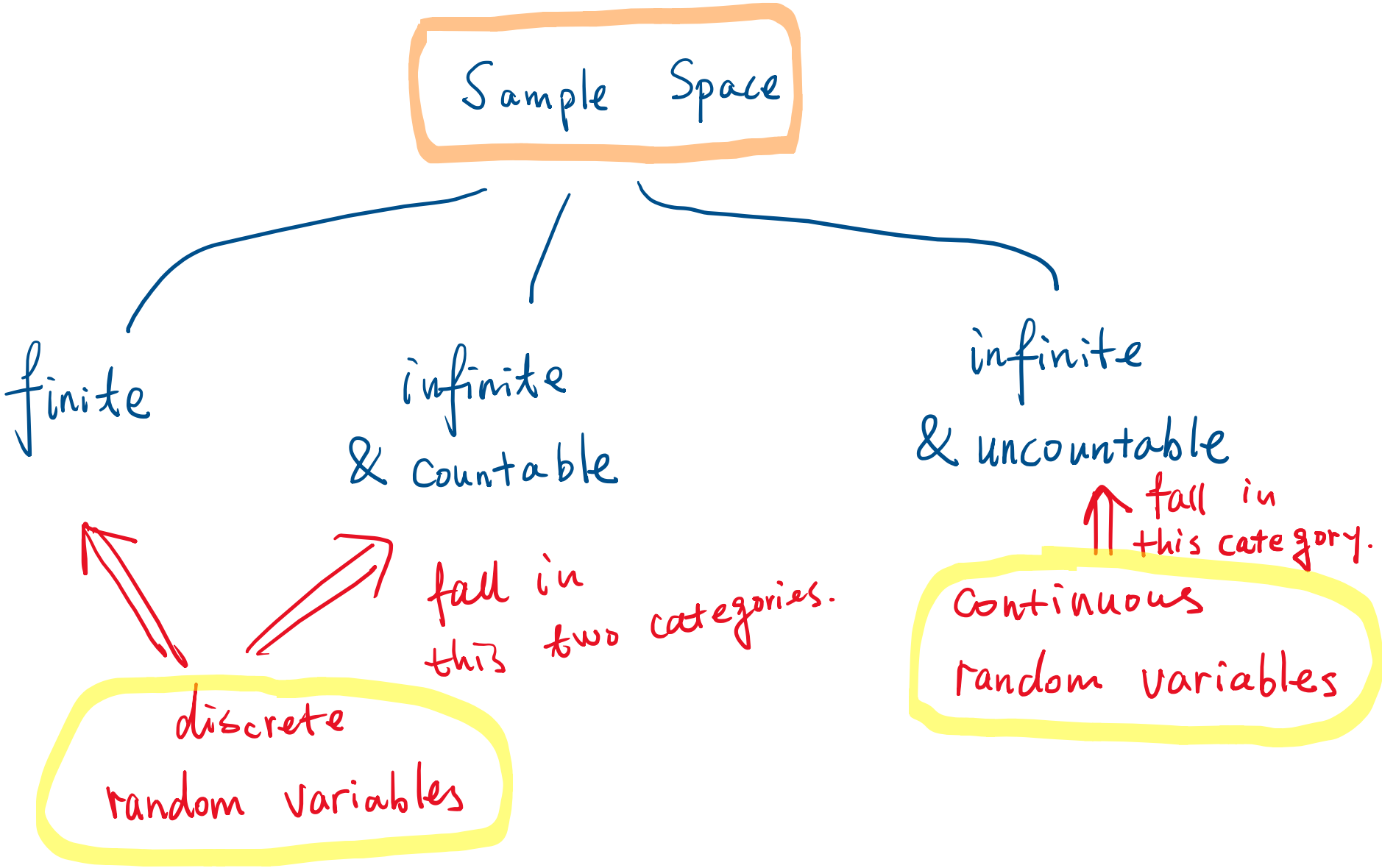
F-distribution

etc.

▷ Pay attention to the concepts,

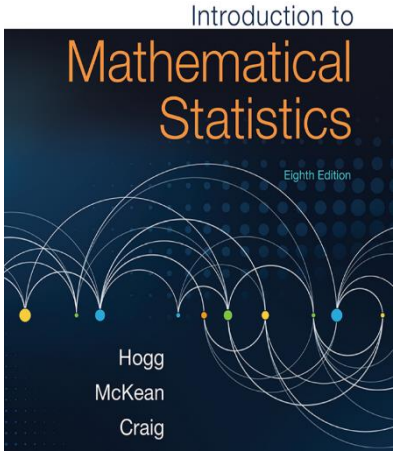
① "sample space"

② "parameter space".


Sample Space
finite
infinite & countable
infinite & uncountable
fall in this two categories.
fall in this category.
discrete random variables
Continuous random variables

Definition 1.6.1 of reference book: 8th Edition

(Discrete Random Variable)



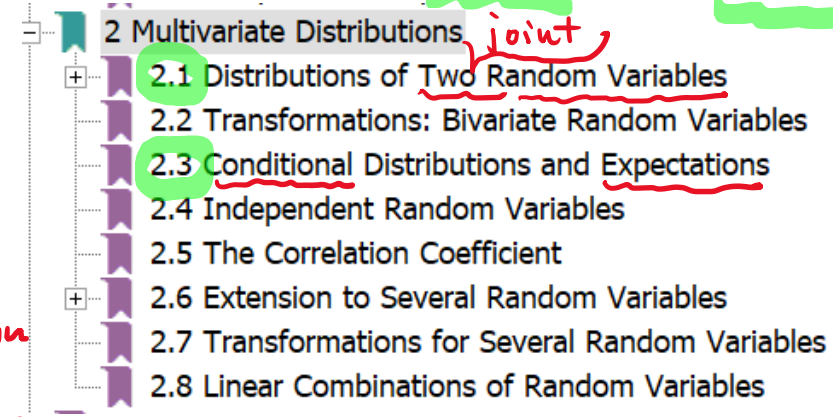
A random variable is a discrete random variable if its sample space is either finite or countable.

Definite 1.7.1 of the reference book.

(Continuous Random Variable)

A random variable is a continuous random variable if its cumulative distribution function (CDF) is continuous for all $x \in \mathbb{R}$.

For discussions on "Multivariate Distributions", the easiest case is "distributions of two random variables". Feel free try more examples and practices, contained in §2-1, §2-3 of the reference book;



Review. ① joint distribution

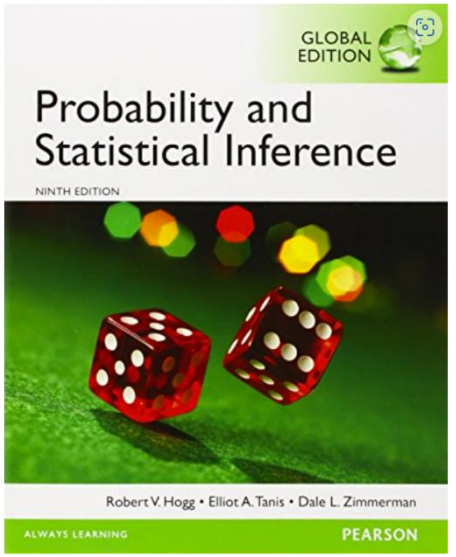
② marginal distribution

③ conditional distribution

④ conditional expectation, first prepare conditional pdf/pmf.

Also feel free to read §4.1 and §4.3 of the textbook (listed in syllabus) thus to review the concepts of "joint / marginal / conditioned distribution" for bivariate distributions.

(two random variables)



4 BIVARIATE DISTRIBUTIONS 125

4.1 Bivariate Distributions of the Discrete Type 125

4.2 The Correlation Coefficient 134

4.3 Conditional Distributions 140

4.4 Bivariate Distributions of the Continuous Type 146

4.5 The Bivariate Normal Distribution 155

Widely use the illustrative examples in the textbook and reference book., to practice and get a better understanding of the concepts. for bivariate.

(two random variables)